

A crash-course on boosting technical skills for writing scientific reports in $\text{\LaTeX}$

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Session content

- ▶ $\text{\LaTeX}$ introduction
- ▶ Useful packages related to:
 - ▶ Text
 - ▶ Math
 - ▶ Graphs
- ▶ References
- ▶ Some good practices for scientific reports
- ▶ Advanced features (version control, Inkscape, ...)
- ▶ Summary

What is this session?

- ▶ It is:
 - ▶ a **technical** session;
 - ▶ a collection of personal experience in form of good practices for better and readable document structure;
 - ▶ emphasis on the improvement of appearance;
 - ▶ a very brief hands-on introduction to $\text{\LaTeX}$.
- ▶ It is NOT:
 - ▶ a session on teaching how to write publications;
 - ▶ a session about the content of the scientific report or manuscript;
 - ▶ university level course.
 - ▶ primary expertise of the speaker (**FG**)

L^AT_EX, the misunderstood monster



Image credits: Wikipedia

Mission of the day: Fall in love with L^AT_EX

- ▶ It makes beautiful documents
 - ▶ Especially mathematics
- ▶ It was created by scientists, for scientists
 - ▶ A large and active community
 - ▶ It is open source
 - ▶ It is standard for STEM¹ publications
- ▶ It is powerful — you can extend it
 - ▶ Packages for papers, presentations, spreadsheets, CVs, . . .
- ▶ Since it is so good, why am I here?

¹Science, Technology, Engineering, and Mathematics

Knowing the audience

1. Have you ever used any programming language?
2. Have you ever used $\text{\LaTeX}$?
3. Have you ever heard of Overleaf?
4. Have you ever used Zotero?
5. Have you ever used version control?

Please, do NOT forget to fill out the survey!

What is $\text{\LaTeX}$?

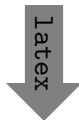
- ▶ It is sort of programming language
- ▶ It is built on top of $\text{\TeX}$, made more user-friendly
- ▶ Collection of predefined macros on $\text{\TeX}$
- ▶ Unlike WYSIWYG² approaches (i.e., MS Word), user focuses on the writing, and "heavy-lifting" is done by $\text{\LaTeX}$
- ▶ Use commands to describe 'what it is', not 'how it looks'.
- ▶ Created by Leslie Lamport
- ▶ Learning curve is initially very steep
 - ▶ This tool is well suited for producing complex, high quality documents

²What You See Is What You Get

How does it work?

- ▶ You write your document in plainb text with **commands** that describe its structure and meaning.
- ▶ The latex program processes your text and commands to produce a beautifully formatted document.

The rain in Spain falls `\emph{mainly}` on the plain.



The rain in Spain falls *mainly* on the plain.

Getting started

- ▶ A minimal $\text{\LaTeX}$ document:

```
\documentclass{article}
\begin{document}
Hello World! % your content goes here...
\end{document}
```

- ▶ Commands start with a *backslash* $\backslash$.
- ▶ Every document starts with a `\documentclass` command.
- ▶ The *argument* in curly braces $\{ \}$ tells $\text{\LaTeX}$ what kind of document we are creating: an *article*.
- ▶ A percent sign $\%$ starts a *comment* — $\text{\LaTeX}$ will ignore the rest of the line.
- ▶ *Small Hello World! via terminal!*

Getting started with **Overleaf**

- ▶ Overleaf is a website for writing documents in $\text{\LaTeX}$.
- ▶ It 'compiles' your $\text{\LaTeX}$ automatically to show you the results.
- ▶ Overleaf projects can be shared with collaborators
- ▶ Premium features offer more advance options (larger number of collaborators, git, ...)
- ▶ Create an Overleaf account
- ▶ Create a new project in Overleaf
- ▶ Explore other templates in Overleaf

Click here to open the example document in **Overleaf**

For best results, please use Google Chrome or a recent FireFox.

- ▶ As we go through the following slides, try out the examples by typing them into the example document on Overleaf.
- ▶ **No really, you should try them out as we go!**

Text

- ▶ Type your text between `\begin{document}` and `\end{document}`.
- ▶ For the most part, you can just type your text normally.

Words are separated by one or more spaces.

Paragraphs are separated by one or more blank lines.

Words are separated by one or more spaces.

Paragraphs are separated by one or more blank lines.

- ▶ Space in the source file is collapsed in the output.

The rain in Spain
falls mainly on the plain.

The rain in Spain falls
mainly on the plain.

Text: Caveats

- ▶ Quotation marks are a bit tricky:
use a backtick ``` on the left and an apostrophe `'` on the right.

Single quotes: ``text'`.

Double quotes: ```text''`.

Single quotes: `'text'`.

Double quotes: `"text"`.

- ▶ Some common characters have special meanings in $\text{\LaTeX}$:

`%`

percent sign

`#`

hash (pound / sharp) sign

`&`

ampersand

`$`

dollar sign

`\`

backslash

`_`

underscore sign

`{`

left brace

`}`

right brace

- ▶ If you just type these, you'll get an error. If you want one to appear in the output, you have to *escape* it by preceding it with a backslash.

`\$ \% \& \# !`

`$ \% \& \# !`

Handling Errors

- ▶ $\text{\LaTeX}$ can get confused when it is trying to compile your document. If it does, it stops with an error, which you must fix before it will produce any output.
- ▶ For example, if you misspell `\emph` as `\meph`, $\text{\LaTeX}$ will stop with an “undefined control sequence” error, because “meph” is not one of the commands it knows.

Advice on Errors

1. Don't panic! Errors happen.
2. Fix them as soon as they arise — if what you just typed caused an error, you can start your debugging there.
3. If there are multiple errors, start with the first one — the cause may even be above it.

Typesetting Exercise 1

Typeset this in $\text{\LaTeX}$: ³

In March 2006, Congress raised that ceiling an additional \$0.79 trillion to \$8.97 trillion, which is approximately 68% of GDP. As of October 4, 2008, the “Emergency Economic Stabilization Act of 2008” raised the current debt ceiling to \$11.3 trillion.

Click to open this exercise in **Overleaf**

- ▶ Hint: watch out for characters with special meanings!
- ▶ Once you’ve tried, [click here to see my solution](#).

³http://en.wikipedia.org/wiki/Economy_of_the_United_States

Title and Abstract

- ▶ Tell $\text{\LaTeX}$ the `\title` and `\author` names in the preamble.
- ▶ Then use `\maketitle` in the document to actually create the title.
- ▶ Use the abstract environment to make an abstract.

```
\documentclass{article}

\title{The Title}

\author{A. Author}

\date{\today}

\begin{document}
\maketitle

\begin{abstract}
Abstract goes here...
\end{abstract}

\end{document}
```

The Title

A. Author

May 19, 2026

Abstract

Abstract goes here...

Sections

- ▶ Just use `\section` and `\subsection`.
- ▶ Can you guess what `\section*` and `\subsection*` do?

```
\documentclass{article}
\begin{document}

\section{Introduction}

The problem of \ldots

\section{Method}

We investigate \ldots

\subsection{Sample Preparation}

\subsection{Data Collection}

\section{Results}

\section{Conclusion}

\end{document}
```

1 Introduction

The problem of ...

2 Method

We investigate ...

2.1 Sample Preparation

2.2 Data Collection

3 Results

4 Conclusion

Labels and Cross-References

- ▶ Use `\label` and `\ref` for automatic numbering.
- ▶ The `amsmath` package provides `\eqref` for referencing equations.

```
\documentclass{article}
\usepackage{amsmath} % for \eqref
\begin{document}
```

```
\section{Introduction}
\label{sec:intro}
```

In Section `\ref{sec:method}`, we `\ldots`

```
\section{Method}
\label{sec:method}
```

```
\begin{equation}
\label{eq:euler}
e^{i\pi} + 1 = 0
\end{equation}
```

By `\eqref{eq:euler}`, we have `\ldots`

```
\end{document}
```

1 Introduction

In Section 2, we ...

2 Method

$$e^{i\pi} + 1 = 0 \tag{1}$$

By (1), we have ...

Structured Documents Exercise

Typeset this short paper in $\text{\LaTeX}$: ⁴

Click to open the paper

Make your paper look like this one. Use `\ref` and `\eqref` to avoid explicitly writing section and equation numbers into the text.

Click to open this exercise in **Overleaf**

- Once you've tried, [click here to see my solution](#).

⁴From <http://pdos.csail.mit.edu/scigen/>, a random paper generator.

Lists

- ▶ The `\begin` and `\end` commands are used to create many different environments.
- ▶ The `itemize` and `enumerate` environments generate lists.

```
\begin{itemize} % for bullet points  
\item Biscuits  
\item Tea  
\end{itemize}
```

- ▶ Biscuits
- ▶ Tea

```
\begin{enumerate} % for numbers  
\item Biscuits  
\item Tea  
\end{enumerate}
```

1. Biscuits
2. Tea

- ▶ For additional example, refer to the *survey.tex*.

Tables

- ▶ Tables in $\text{\LaTeX}$ take some getting used to.
- ▶ Recent updates offer GUI for building tables.
- ▶ Use the `tabular` environment from the `tabularx` package.
- ▶ The argument specifies column alignment — **left, right, right**.

```
\begin{tabular}{lrr}  
Item & Qty & Unit $ \\  
Widget & 1 & 199.99 \\  
Gadget & 2 & 399.99 \\  
Cable & 3 & 19.99 \\  
\end{tabular}
```

Item	Qty	Unit \$
Widget	1	199.99
Gadget	2	399.99
Cable	3	19.99

- ▶ It also specifies vertical lines; use `\hline` for horizontal lines.

```
\begin{tabular}{|l|r|r|} \hline  
Item & Qty & Unit $ \\\hline  
Widget & 1 & 199.99 \\  
Gadget & 2 & 399.99 \\  
Cable & 3 & 19.99 \\\hline  
\end{tabular}
```

Item	Qty	Unit \$
Widget	1	199.99
Gadget	2	399.99
Cable	3	19.99

- ▶ Use an ampersand `&` to separate columns and a double backslash `\\` to start a new row (like in the `align*` environment that we saw in part 1).

Codelistings

- ▶ Useful for writing user-manuals.
- ▶ Handy for highlighting, embedding code snippets.
- ▶ Preserves formatting
- ▶ Refer to *matlab_code_example.tex* and *python_example.tex*
- ▶ L^AT_EX snippets on this presentation make use of verbatim package

Command customization

- It is handy to parameterize existing `\commands` by re-defining them using `\newcommand`

```
\documentclass{article}
\include{new_command_preamble}
\begin{document}
\newcommand{\bb}[1]{\mathbb{#1}}
The complex numbers:\(\ \bb{C} \),\,
and the integer numbers \(\ \bb{Z} \).\,
\newcommand{\plusbin}[4][2]{(#2 + #3)^{#1}}
We can use it like this:
\[\ \plusbin{x}{y}{5} \]
And even the exponent can be changed:
\[\ \plusbin[4]{a}{b}{g} \]
The standard syntax for defining a command
\\ with an optional default value is:
\begin{lstlisting}
\newcommand{\commandname}[number of arguments]
[default value]{definition}
\end{lstlisting}
\end{document}
```

The complex numbers: $\mathbb{C}$,

and the integer numbers $\mathbb{Z}$.

We can use it like this:

$$(x + y)^{2+}$$

And even the exponent can be changed:

$$(a + b)^{4+!}$$

The standard syntax for defining a command with an optional default value is:

```
\newcommand{\commandname}[number of arguments]
[default value]{definition}
```

Typesetting Mathematics: Dollar Signs

- Why are dollar signs $\$$ special? We use them to mark mathematics in text.

% not so good:

Let a and b be distinct positive integers, and let $c = a - b + 1$.

% much better:

Let a and b be distinct positive integers, and let $c = a - b + 1$.

Let a and b be distinct positive integers, and let $c = a - b + 1$.

Let a and b be distinct positive integers, and let $c = a - b + 1$.

- Always use dollar signs in pairs — one to begin the mathematics, and one to end it.
- $\LaTeX$ handles spacing automatically; it ignores your spaces.

Let $y=mx+b$ be $\ldots$

Let $y = m x + b$ be $\ldots$

Let $y = mx + b$ be $\ldots$

Let $y = mx + b$ be $\ldots$

Typesetting Mathematics: Notation

- Use caret `^` for superscripts and underscore `_` for subscripts.

```
$y = c_2 x^2 + c_1 x + c_0$
```

$$y = c_2 x^2 + c_1 x + c_0$$

- Use curly braces `{}` `}` to group superscripts and subscripts.

```
$F_n = F_{n-1} + F_{n-2}$ % oops!
```

$$F_n = F_n - 1 + F_n - 2$$

```
$F_n = F_{n-1} + F_{n-2}$ % ok!
```

$$F_n = F_{n-1} + F_{n-2}$$

- There are commands for Greek letters and common notation.

```
$\mu = A e^{Q/RT}$
```

$$\mu = A e^{Q/RT}$$

```
$\Omega = \sum_{k=1}^n \omega_k$
```

$$\Omega = \sum_{k=1}^n \omega_k$$

Typesetting Mathematics: Displayed Equations

- If it's big and scary, *display* it on its own line using `\begin{equation}` and `\end{equation}`.

The roots of a quadratic equation
are given by
`\begin{equation}`
`x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}`
`\end{equation}`
where `$a$`, `$b$` and `$c$` are `\ldots`

The roots of a quadratic
equation are given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (1)$$

where a , b and c are ...

Caution: $\text{\LaTeX}$ mostly ignores your spaces in mathematics, but it can't handle blank lines in equations — don't put blank lines in your mathematics.

Interlude: Environments

- ▶ `equation` is an *environment* — a context.
- ▶ A command can produce different output in different contexts.

We can write

```
$ \Omega = \sum_{k=1}^n \omega_k $
```

in text, or we can write

```
\begin{equation}
```

```
\Omega = \sum_{k=1}^n \omega_k
```

```
\end{equation}
```

to display it.

We can write $\Omega = \sum_{k=1}^n \omega_k$
in text, or we can write

$$\Omega = \sum_{k=1}^n \omega_k \quad (2)$$

to display it.

- ▶ Note how the Σ is bigger in the `equation` environment, and how the subscripts and superscripts change position, even though we used the same commands.

In fact, we could have written `$...$` as `\begin{math}...\end{math}`.

Interlude: Environments

- ▶ Usage of matrices and equations
- ▶ Refer to *matrices_example*

```
\documentclass{beamer}
\begin{document}
\begin{align}
\mathbf{T}^{1}_{0} &= \begin{bmatrix}
c_{\theta_1} & 0 & s_{\theta_1} & 0 \\
s_{\theta_1} & 0 & -c_{\theta_1} & 0 \\
0 & 1 & 0 & d_1 \\
0 & 0 & 0 & 1
\end{bmatrix}.
\end{align}
\end{document}
```

- ▶ For advanced usage of matrices, and equations, check *equation.tex*

Typesetting Mathematics: Examples with `amsmath`

- Use `equation*` (“equation-star”) for unnumbered equations.

```
\begin{equation*}
\Omega = \sum_{k=1}^n \omega_k
\end{equation*}
```

$$\Omega = \sum_{k=1}^n \omega_k$$

- $\text{\LaTeX}$ treats adjacent letters as variables multiplied together, which is not always what you want. `amsmath` defines commands for many common mathematical operators.

```
\begin{equation*} % bad!
\min_{x,y} (1-x)^2 + 100(y-x^2)^2
\end{equation*}
\begin{equation*} % good!
\min_{x,y} \{(1-x)^2 + 100(y-x^2)^2\}
\end{equation*}
```

$$\min_{x,y} (1-x)^2 + 100(y-x^2)^2$$

$$\min_{x,y} \{(1-x)^2 + 100(y-x^2)^2\}$$

- You can use `\operatorname` for others.

```
\begin{equation*}
\beta_i =
\frac{\operatorname{Cov}(R_i, R_m)}
{\operatorname{Var}(R_m)}
\end{equation*}
```

$$\beta_i = \frac{\operatorname{Cov}(R_i, R_m)}{\operatorname{Var}(R_m)}$$

Typesetting Mathematics: Examples with `amsmath`

- Align a sequence of equations at the equals sign

$$\begin{aligned}(x+1)^3 &= (x+1)(x+1)(x+1) \\ &= (x+1)(x^2+2x+1) \\ &= x^3+3x^2+3x+1\end{aligned}$$

with the `align*` environment.

```
\begin{align*}
(x+1)^3 &=& (x+1)(x+1)(x+1) &\\
&=& (x+1)(x^2+2x+1) &\\
&=& x^3+3x^2+3x+1 &\\
\end{align*}
```

- An ampersand `&` separates the left column (before the `=`) from the right column (after the `=`).
- A double backslash `\\` starts a new line.

Typesetting Exercise 2

Typeset this in $\text{\LaTeX}$:

Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed random variables with $E[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2 < \infty$, and let

$$S_n = \frac{1}{n} \sum_{i=1}^n X_i$$

denote their mean. Then as n approaches infinity, the random variables $\sqrt{n}(S_n - \mu)$ converge in distribution to a normal $N(0, \sigma^2)$.

Click to open this exercise in **Overleaf**

- ▶ Hint: the command for ∞ is `\infty`.
- ▶ Once you've tried, [click here to see my solution](#).